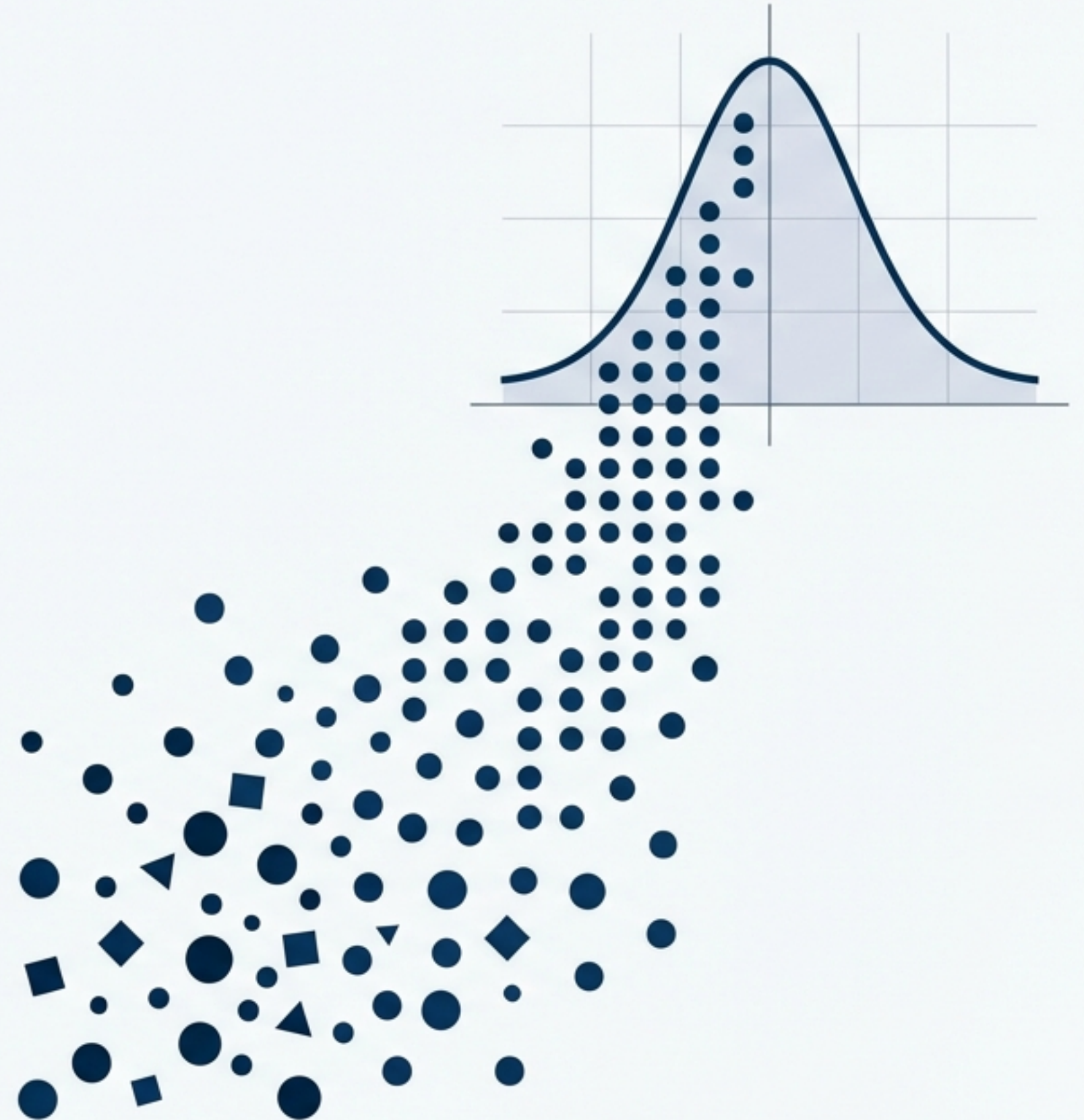


# Statistical Decision Making: From Uncertainty to Verdict

A guide to validating data insights using Hypothesis Testing, P-Values, and Confidence Intervals.

Roboto

PYTHON IMPLEMENTATION & REAL-WORLD APPLICATION





# The Core Challenge: Signal vs Noise

Inter Tight

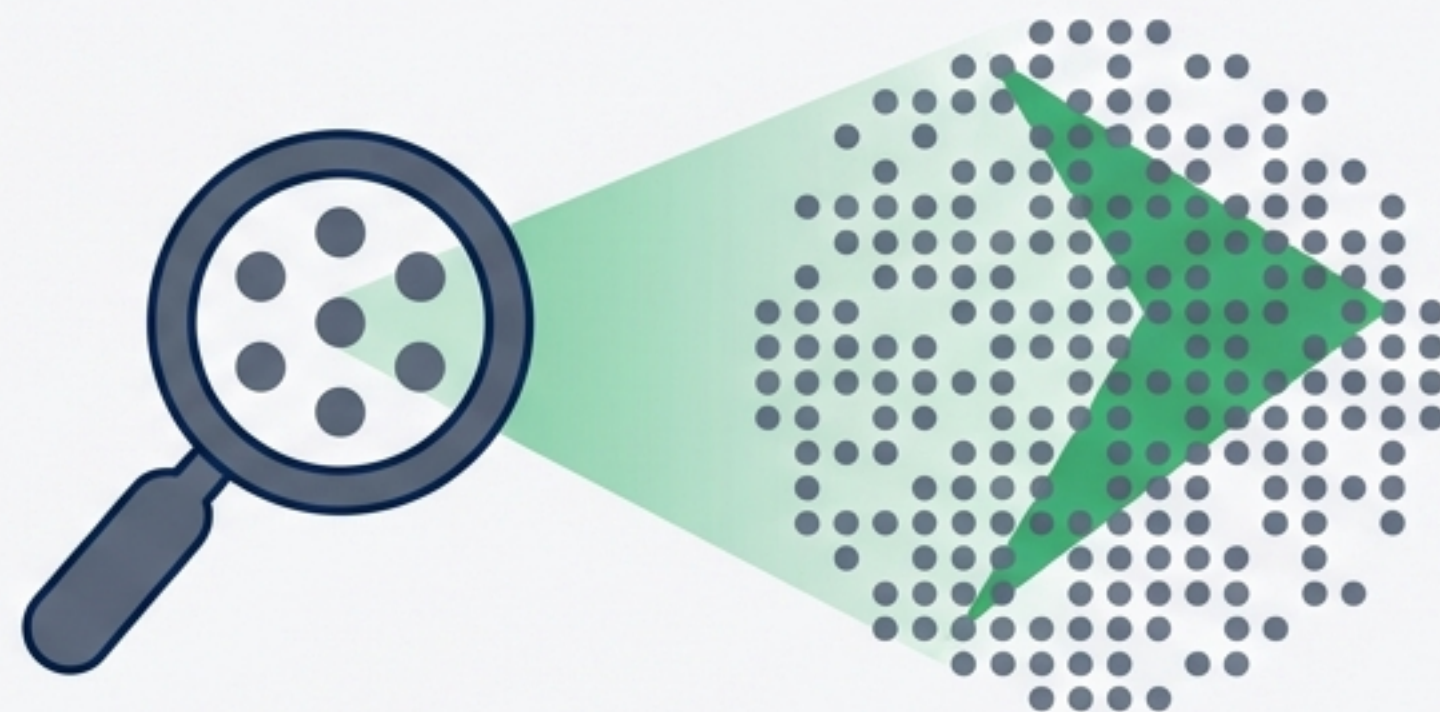
## Probability (Chance)



The chance of a single event happening.  
E.g., Coin toss = 50%.

Inter Tight

## Inferential Statistics (Insight)



Using sample data to make valid claims  
about a whole population.

**The Goal:** Moving from '**it looks different**' to '**it is statistically different**'.

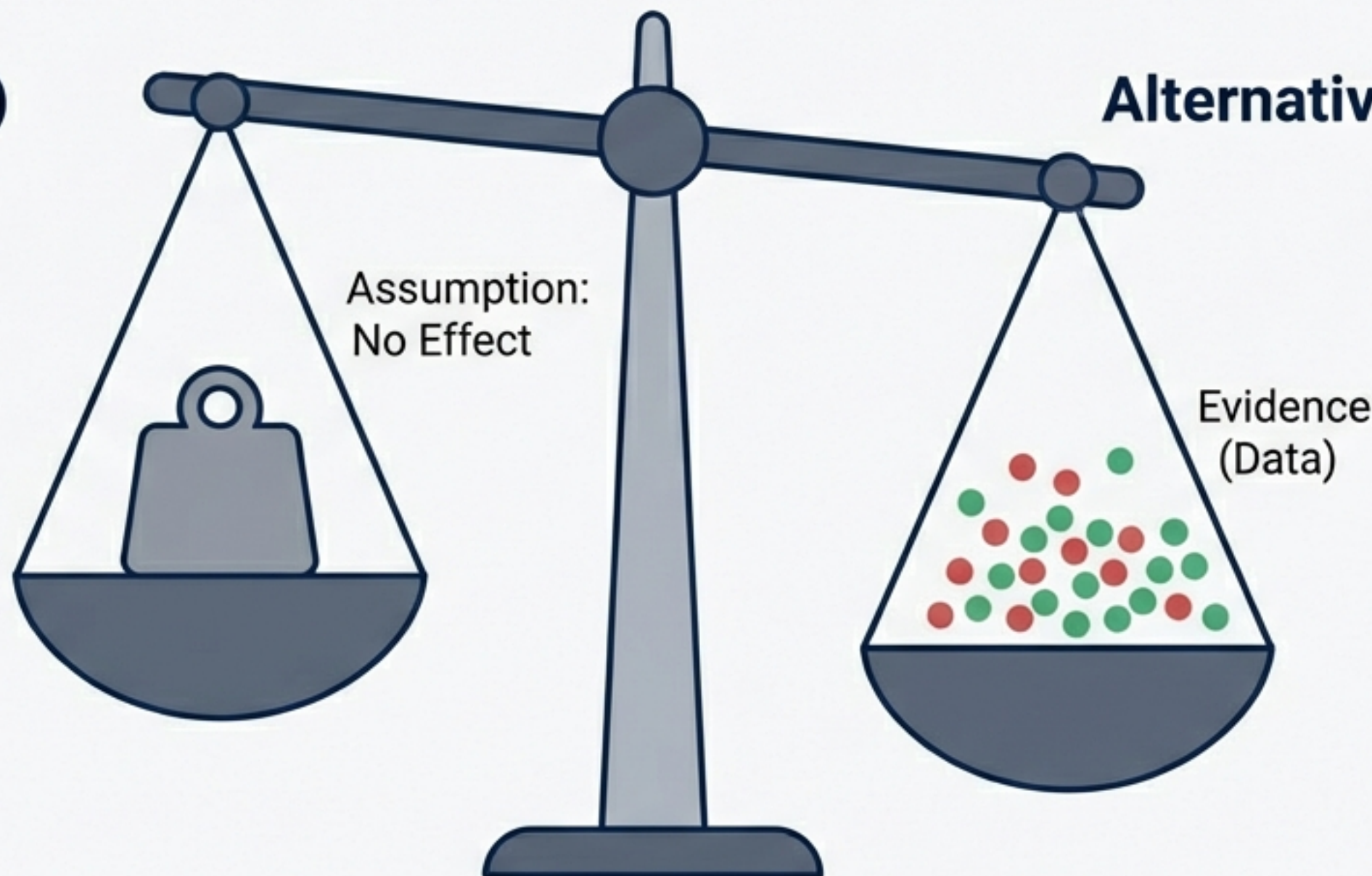


# Hypothesis Testing: The 'Trial' of Data

Inter Tight

**Null Hypothesis ( $H_0$ )**

$H_0$  is Innocent until proven guilty. We assume there is no difference (e.g. "The drug has no effect").



Inter Tight

**Alternative Hypothesis ( $H_1$ )**

$H_1$  is the claim we want to prove. We assume there IS a difference (e.g. "The drug lowers blood pressure").

**The Rule:** We hold  $H_0$  as true unless the data is so weighty that we are forced to reject it.



# The Evidence: What is a P-Value?

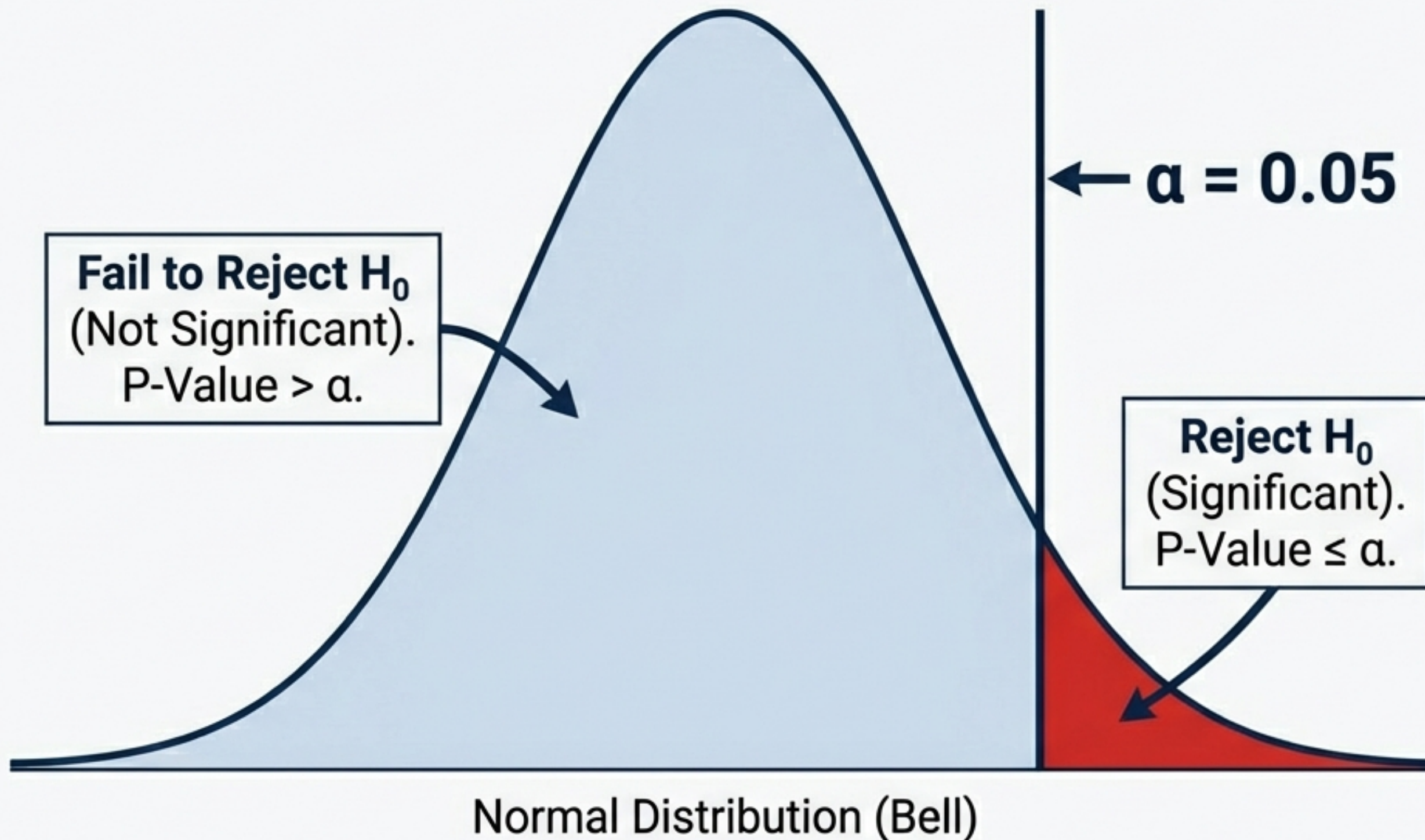
A measure of surprise, not truth.



- Definition: The probability of observing these results if  $H_0$  were true.
- **CRITICAL**: It is **NOT** the probability that  $H_0$  is true.
- Rule: **Lower P-value = Stronger evidence** against the Null Hypothesis.



# The Threshold: Significance Value ( $\alpha$ )



Significance Level ( $\alpha$ ):  
The 'Cut-off' point for  
decision making.

Standard Standards:

- 0.05 (5%) for 95% Confidence
- 0.01 (1%) for 99% Confidence



# Execution: Running a T-Test in Python

**Scenario:** Factory claim is 50g. Sample mean is 48g. Is the difference real?

```
code.py x

import scipy.stats as stats

# Sample weights (Mean = 48)
sample = [49, 50, 48, 47, 51, 50, 49, 48, 47, 48]

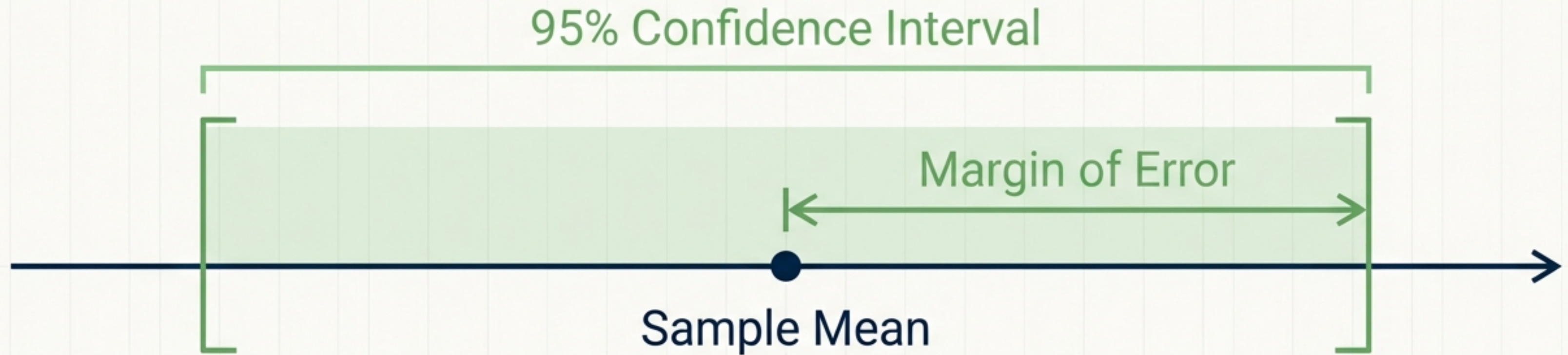
# Perform t-test against population mean of 50
t_stat, p_value = stats.ttest_1samp(sample, 50)
print(f"T-statistic: {t_stat}")
print(f"P-value: {p_value}")
```

```
Output:
T-statistic: -2.345207889194903
P-value: 0.041
```

Value is  $< 0.05$ .  
Decision: Reject Claim.



# Confidence Intervals: The Range of Truth



$$CI = \text{Sample Mean} \pm (\text{Critical Value} \times \text{Standard Error})$$

- Hypothesis testing asks “Is there a difference?” (**Yes/No**).
- Confidence Intervals ask “Where does the true value likely lie?” (**Range**).



# Execution: Calculating the Confidence Interval

```
code.py x

import numpy as np
import scipy.stats as stats

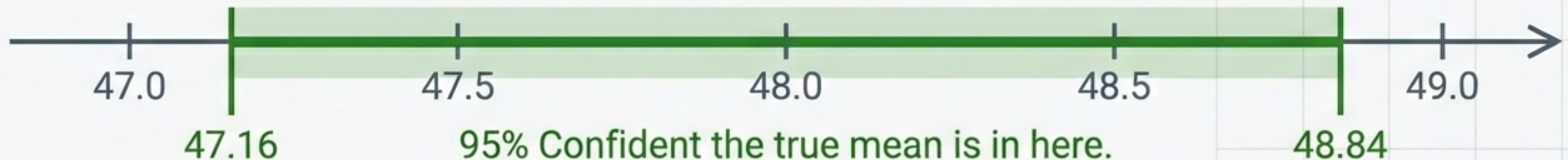
data = [49, 50, 48, 47, 51, 50, 49, 48, 47, 48]
mean = np.mean(data)
std = np.std(data, ddof=1)
n = len(data)

# Calculate 95% CI
z = stats.t.ppf((1 + 0.95) / 2, df=n-1)

# Calculate 95% CI
margin_of_error = z * (std / np.sqrt(n))

lower = mean - margin_of_error
upper = mean + margin_of_error

print(f"95% CI: [{lower:.2f}, {upper:.2f}]" )
```





# When Decisions Go Wrong: Type I & Type II Errors

|                            | Reality is $H_0$ True                                                                                                  | Reality is $H_0$ False                                                                                                   | No statistical test is perfect.<br><br>We weigh the risk of a False Alarm against the risk of a Missed Opportunity. |
|----------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Decision<br>Reject $H_0$   | <b>Type I Error (False Alarm)</b><br><br>Rejecting $H_0$ when it is actually true.<br>(Convicting an innocent person). | <b>Correct Decision (True Positive)</b>                                                                                  |                                                                                                                     |
| Decision<br>Fail to Reject | <b>Correct Decision (True Negative)</b>                                                                                | <b>Type II Error (Missed Opportunity)</b><br><br>Failing to reject $H_0$ when it is false. (Letting the guilty go free). |                                                                                                                     |



# Case Study 1: The Bottle Filling Machine

**Scenario:** A machine claims to fill 500ml. We sample 30 bottles.

## The Hypotheses

- $H_0: \mu = 500$  (Machine is working)
- $H_1: \mu \neq 500$  (Machine is malfunctioning)

## The Simulation

```
np.random.seed(42)  
# Simulating 30 bottles with a mean of 498ml  
data = np.random.normal(loc=498, scale=3, size=30)
```





# Case Study 1: The Verdict

```
stats.ttest_1samp(data, popmean=500)
```

---

**T-statistic: -3.842**

**P-value: 0.001**

**Decision: REJECT  $H_0$**

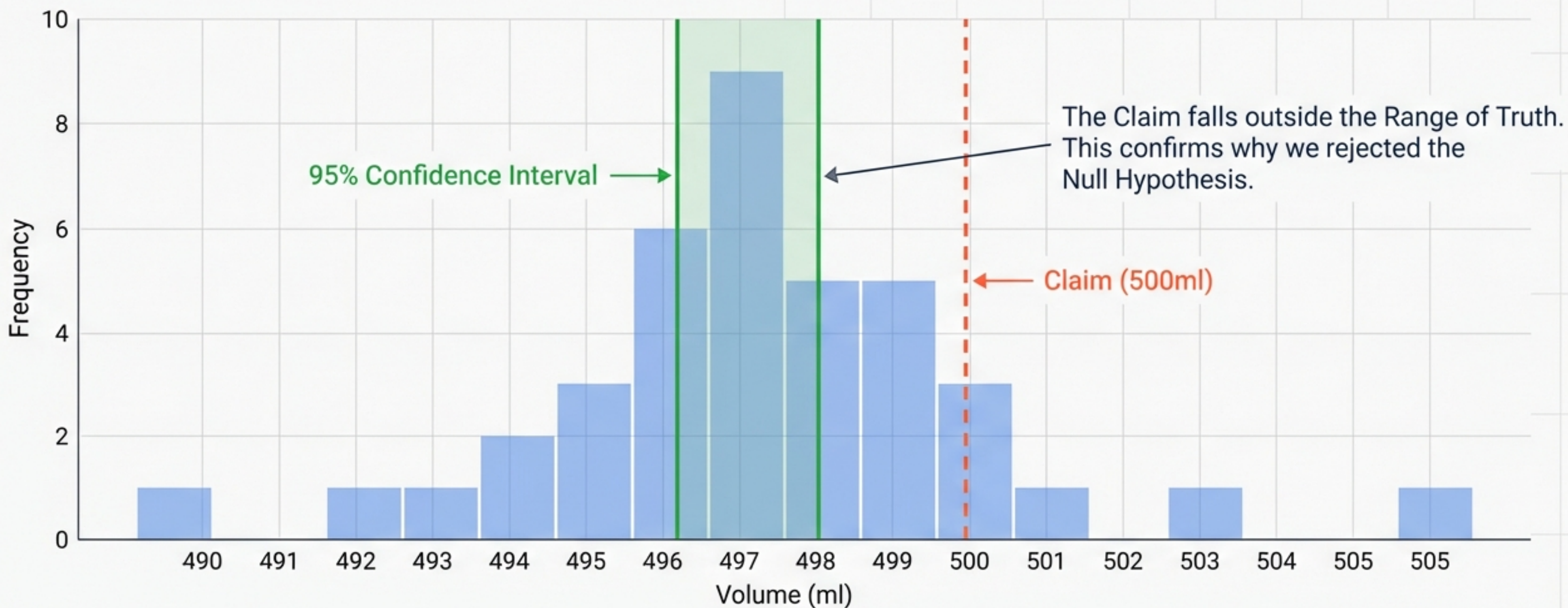
**Reason:** P-value (0.001) < Alpha (0.05).

**Conclusion:** The machine is statistically confirmed to be underfilling.



# Case Study 1: Visualising the Evidence

Comparing the Claim against the Reality.





# Case Study 2: The New Drug Trial

Real-world consequences of statistical decisions. Roboto

| The Facts                                      | Risk of Type I Error                                                                 | Risk of Type II Error                                                                            |
|------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Hypothesis:</b> Drug lowers blood pressure. |  |              |
| <b>Result:</b> P-value = 0.03.                 | If we are wrong ( <b>False Positive</b> ):<br>We sell a useless drug to patients.    | If we had missed it ( <b>False Negative</b> ):<br>We scrap a working cure that could save lives. |
| <b>Verdict:</b> Reject $H_0$ . The drug works. |                                                                                      |                                                                                                  |

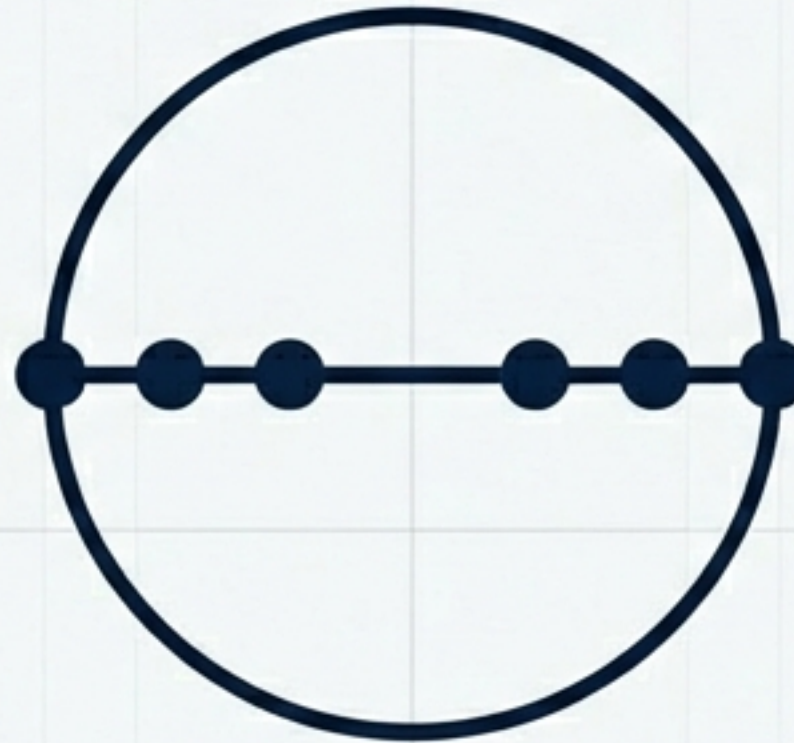


# The Statistical Toolkit: Summary

| Concept             | Function                           | Key Metric                          | Decision Rule                              |
|---------------------|------------------------------------|-------------------------------------|--------------------------------------------|
| Hypothesis Testing  | Formal process to test assumptions | T-statistic                         | Compare Sample vs. Population              |
| P-Value             | Measures evidence against $H_0$    | Probability (0 to 1)                | If < Alpha, <b>Reject <math>H_0</math></b> |
| Alpha ( $\alpha$ )  | Decision Threshold                 | <b>0.05 (Standard)</b>              | Sets the strictness                        |
| Confidence Interval | Estimates true population range    | <b>Mean <math>\pm</math> Margin</b> | Does the <b>range include</b> the claim?   |



# Conclusion: The Rules of Evidence



**“Data provides the evidence. Statistics provides the rules. You make the verdict.”**

Statistics does not provide absolute certainty; it provides a framework to minimise error in our decision-making.